VU Medical Zone

BT 603 Current Final Term Updated Spring 2021

1. Fermentation can be classified into _____ types. **7**

2. On the basis of Oxygen fermentation are classified on different types? **3**

3. Molasses is converted into Alcohol with the help of_____. **Yeast**

4. On the basis of scale, there are following types of fermentation? **3**

5. **Selective media** are designed to suppress the growth of unwanted bacteria and encourage the growth of the desired microbes.

6. Many media are buffered at about pH **7.0** by the incorporation of calcium carbonate.

7. Many antifoams are of **low solubility** they need a carrier such as lard oil, liquid paraffin or castor oil.

8. The optimum range of **osmotic pressure** for growth is often quite narrow and varies with the type of cell and the species from which it was isolated.

9. The **foaming** can cause removal of cells from the medium which will lead to autolysis and the further release of microbial cell proteins.

10. Nutrient, antifoam, pH and temperature are_____. **Independent variable**

11. When more than five independent variables are to be investigated, the **Plackett-Burman** design may be used.

12. The oxygen demand of an industrial fermentation process is normally satisfied by **aerating** the fermentation broth.

13. The solubility of oxygen in pure water is_____. **8 mg/L at 4oC**

14. The rate of oxygen transfer from air bubble to the liquid phase may be described by the equation_____. **$dC_L / dt = K_L a (C^* - C_L)$**

15. The units of K_La is_____. **h^{-1}**

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16. Streak plate method for pure culturing of required microorganism is done for_____. **Non-filamentous unicellular microorganisms**

17. **Strain improvement** involves the continual genetic modification of the culture, followed by reappraisals of its cultural requirements.

18. A large-scale continuous fermentation for the production of biomass would be expected to operate at steady state in excess of **100** days.

19. Lincoln _____ suggested a more elaborate procedure for the development of inoculum. **1960**

Subjective

Q1. What is the function of Impeller in the fermentation process? 2 Marks

Fermentation involves proper mixing during the process of fermentation which is done by Agitator/ Impeller.

Q2. Name any Four inhibitor used in fermentation. 2 Marks

1. Sodium bisulphate
2. Bromide
3. Penicillin
4. Di-ethyl barbiturate

Q3. How oxygen is supplied to the microbial culture during fermentation? 5 Marks

Oxygen Supply

Oxygen is normally supplied to microbial cultures in the form of air, this being the cheapest available source of the gas. The method for provision of a culture with a supply of air varies with the scale of the process:

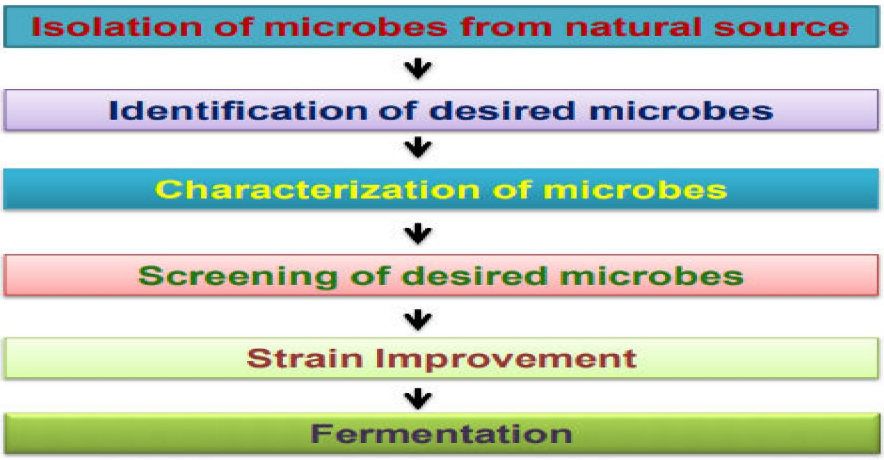
Laboratory-scale cultures may be aerated by means of the shake-flask technique where the culture (50 to 100 cm3) is grown in a conical flask (250 to 500 cm3) shaken on a platform contained in a controlled environment chamber.

Pilot- and industrial-scale fermentations are normally carried out in stirred, aerated vessels (fermenters).

Q4. Dynamic and static method is suitable for the determination of K_La value? 2 Marks

Both the dynamic and static methods are also unsuitable for measuring KLa values in viscous systems.

Q5. Explain INDUSTRIAL STRATEGY FOR USAGE OF MICROORGANISMS with Flow sheet Diagram? 5 Marks



Q6. Explain the Antibiotic Screening test? 5 Marks

The evolution of antibiotic screens serves as an excellent illustration of the development of more precise, targeted systems.

Antibiotics were initially detected by growing the potential producer on an agar plate in the presence of an organism (or organisms) against which antimicrobial action was required.

Production of the antibiotic was detected by inhibition of the test organism(s).

Alternatively, the microbial isolate could be grown in liquid culture and the cell-free broth tested for activity.

This approach was extended by using a range of organisms to detect antibiotics with a defined antibacterial spectrum.

Q7. Name the various Phenotypic technique for identification of Microorganism? 3 Marks

• Microscopic morphology

• Metabolic differences (Biochemical Testing)

• Serology

• Fatty acid analysis

Q8. Write down the Types of Physical Mutagens. 2 marks

UV and the Ionizing radiations: X-rays, gamma rays, alpha particles and fast neutrons.

Q9. Methods for genetic technique industrial process. 3 marks

1. Conventional mutation/Random Mutagenesis

2. Transduction

3. Conjugation

4. Transformation

6. Protoplast fusion

7. Genetic engineering

8. Metabolic engineering

9. Site-directed mutagenesis

Q10. Explain Ames Test. 5 marks

The Ames test is a relatively inexpensive and rapid test for identifying possible chemical carcinogens. The test assumes that a mutant cell can revert to a normal cell in the presence of a mutagen and that many mutagens are carcinogens.

Histidine auxotrophs (require histidine in the media) of Salmonella are exposed to an enzymatically treated potential carcinogen, and reversions to the nonmutant state are selected.

Before considering the methods used for the selection of methods producing improved levels of primary metabolites it is necessary to study the mechanisms of control of their biosynthesis such that the 'blueprints', may be drawn accurately.

Q11. Positive and Negative selection of mutant? 3 Marks

Positive selection involves the selection of mutant cells and rejection of nonmutated cells. To identify mutants that have gained some physical trait.

Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait – for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

Q12. Henry's Law? 2 Marks

Henry's Law describes the solubility of O2 in nutrient solution in relation to the O2 partial pressure in the gas phase.

$$C^* = P_0 / H$$

C* is the oxygen saturation concentration of the nutrient solution,

Po is the partial pressure of the gas in the gas phase

H is Henry's constant, which is specific for the gas and the liquid phase.

Q13. Name the compounds used in Antifoaming? 5 marks

1. Alcohols; stearyl and octyl decanol.

2. Esters.

3. Fatty acids and derivatives, particularly glycerides, which include cottonseed oil, linseed oil, say-bean oil, olive oil, castor oil, sunflower oil, rapeseed oil and cod liver oil.

4. Silicones.

5. Sulphonates.

6. Miscellaneous; Alkaterge C, oxazaline, poly-propylene glycol.

Q14. Write two examples of methylotrophic yeast? 2 marks

Methylotrophic yeasts such as Hansenula polymorpha and Pichia pastoris.

Q15. Write three ways for control foaming? 3 Marks

1. To try and avoid foam formation by using a defined medium and a modification of some of the physical parameters (pH, temperature, aeration and agitation). This assumes that the foam is due to a component in the medium and not a metabolite.

2. The foam is unavoidable and antifoam should be used. This is the more standard approach.

3. To use a mechanical foam breaker.

Q16. Name the five cultures collection. 5 Marks

1. National Collection of Type cultures (NCTC)

2. National Collection of Yeast Cultures (NCYC)

3. American Type Culture Collection (ATCC)

4. Collection of International Mycological Institute (IMl)

5. Culture Collection of the Institute for Fermentation (IFO)

Q17. Write down the factors affecting K_La value. 3 Marks

Factors affecting KLa the air-flow rate employed in vessels,

–the degree of agitation inside vessels,

– the rheological properties of the culture broth and

– the presence of antifoam agents.

Q18. Formation of Protoplasts? 2 Marks

Protoplasts are formed from bacteria, fungi, yeast and actinomycetes when dividing cells are caused to lose their cell walls.

Q19. Problems in Medium optimization? 2 Marks

Medium optimization by the classical method of changing one independent variable (nutrient, antifoam, pH, temperature, etc.) while fixing all the others at a certain level can be extremely time consuming and expensive for a large number of variables.

Q20. Chemical Mutagens? 3 Marks

Chemical mutagens (alkylating agents, base analogues, deaminating agents) could be used for increasing the mutation rate of bacteria. Then we can select the ones with desired properties.

Q21. Steps involved in transferring oxygen from air to cell in fermentation? 3 Marks

Bartholomew et al. (1950) represented the transfer of oxygen from air to the cell, during a fermentation, as occurring in a number of steps:

i. The transfer of oxygen from an air bubble into solution.

ii. The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.

iii. The uptake of the dissolved oxygen by the cell.

Q22. Aeration & Agitation? 5 Marks

The majority of fermentation processes are aerobic and, therefore, require the provision of oxygen. If the stoichiometry of respiration is considered, then the oxidation of glucose. However, both components must be in solution before they are available to a micro-organism and oxygen is approximately 6000 times less soluble in water than is glucose.

A fermentation medium saturated with oxygen contains approximately 7.6 mg dm-3 of oxygen at 30°C. Thus, it is not possible to provide a microbial culture with all the oxygen it will need for the complete oxidation of the glucose (or any other carbon source) in one addition. Therefore, a microbial culture must be supplied with oxygen during growth at a rate sufficient to satisfy the organisms' demand. The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth.

Q23. CPR? 5 Marks

Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as a transfer criterion which requires analysis of the fermenter effluent air. This approach is suitable only when transfer is being made from a fermenter. These workers provide an excellent example of the effect of inoculum transfer time on the productivity of a streptomycete secondary metabolite. The carbon dioxide production rate (CPR) profile of the inoculum culture showing the points 0, 2 and 3) at which inocula were removed. The effect of inoculum age on the CPR of the production fermentation, it may be seen that the three fermentations performed similarly. Thus, although the time of transfer had only a marginal influence on biomass in the production fermentation the effect on product formation was critical.